

Ch 2 questions

1. Potential outcomes notation:

- a. Outcomes if subject is not exposed to treatment
- b. d_i indicates a treatment that a subject receives. D_i is a treatment that can be hypothetically administered. Both indicate i th subject.
- c. $Y_i(0)$ only indicates that the subject is assigned to not be exposed to treatment. $Y_i(0)|D_i=0$ indicates that the subject was assigned to not be exposed, and they were not administered treatment.
- d. $Y_i(0)|D_i=1$ indicates that the subject was assigned to not be exposed, and they were administered treatment. $Y_i(0)|D_i=0$ indicates that the subject was assigned to not be exposed, and they were not administered treatment.
- e. $E[Y_i(0)]$ refers to the expected value of $Y_i(0)$ when one subject is sampled at random. $E[Y_i(0)|D_i=1]$ refers to the expected value of $Y_i(0)|D_i=0$ when one subject is sampled at random from those who were not hypothetically treated.
- f. The selection bias term is zero due the fact that there would not be $Y_i(0)|D_i=1$ as no one received treatment that was not assigned to them.

2. Student newspaper

- a. This may jeopardise the excludability assumption as it is not only the treatment of the newspaper that might cause a difference between the treatment and control groups, but also the knowledge by the treatment group that they are part of a study.
- b. This may jeopardize non-interference assumptions as students that are part of the control condition could be affected by the newspapers brought by the treatment condition. The control students might read the newspaper during lunch.

3. Inmates

- a. We would be hesitant to assume anything since the conditions are not randomly assigned. We only find correlations between reading and violence as inmates self select into conditions.
- b. This study design has multiple treatments going on at the same time, which causes a question about the excludability assumption. The treatment group are also going

to the prison library and forced to be in specially designated carrels for three hours. This in of itself may cause behaviour changes.

- c. The assumption of non-interference is that the behaviour of one inmate does not change behaviour of another inmate.
- d. The non-interference assumption states that every individual's behaviour is only in response to a policy where all prisoners are required to read for three hours, not the behaviour of other inmates.